Hardware-Validated Quantum-Classical Feedback Loops for Cognitive Architectures: The Quantum Organic Intelligence (QOI) Framework

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ABSTRACT

We present the Quantum Organic Intelligence (QOI) framework: a hardware-validated quantum-classical feedback system coupling a Gross-Pitaevskii fluid solver (16 cognitive modes) with a 2-qubit quantum bridge on IBM 156-qubit superconducting QPUs. The system implements three separate quantum measurements (A, f, φ) per iteration, with real-time Bell-fidelity anchoring and closed-loop decoding via a Rust bridge. We report 108 hardware iterations (68+20+20) on `ibm_fez` under the IBM Quantum open plan (600 s quota, no Runtime Sessions), measuring queue wait (17.9 s/iter), QPU time (2.96 s/iter), mode distributions, and per-channel decoding accuracy. The quantum path achieves 0% mode-argmax accuracy vs. 100% classical control; the c-indicator (Bell balance) tracks hardware drift but fails as a decoding correction due to asymmetric readout noise. Complementary validation: Quantum Delayed Synchronization (QDS) Hahn-echo $T_2 = 89.6 \pm 39.2\ \mu$s; Quantum Anchor protocol for coherent-state anchoring; Wave-Quantum frequency-to-correlation encoding (Pearson 0.87–0.999); Quantum Anchored Memory (QAM) integrating $\psi(t)$ dynamics with a 9-level zoom microscope memory (Merkle integrity, spine bus). We release all raw data (104 measurement files), analysis scripts, and open-source crates (`fluid-stratos`, `quantum-bridge`) under MIT license.

CCS CONCEPTS

• **Computer systems organization** → **Quantum computing**; *Quantum error mitigation*; *Quantum algorithms*; • **Computing methodologies** → **Cognitive architectures**.

KEYWORDS

Quantum-classical feedback, Gross-Pitaevskii equation, superconducting QPU, hardware validation, neuromorphic memory, quantum error mitigation

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1 INTRODUCTION

Quantum-classical hybrid algorithms promise advantage by delegating structured subroutines to quantum hardware while retaining classical control. Yet most demonstrations remain in simulation or use shallow circuits without sustained feedback. We ask: *can a quantum-classical loop run on real hardware long enough to exhibit dynamical structure, and what does the hardware teach us about the loop's viability?*

The Quantum Organic Intelligence (QOI) framework answers this by integrating:

(1) **Fluid STRATOS**: a 2D Gross-Pitaevskii equation (GPE) solver with real split-step Fourier kinetics (`rustfft 6.4.1`, Strang splitting), hosting 16 cognitive modes as attractors.
(2) **Quantum Bridge**: 2 qubits, 3 separate measurements per iteration — amplitude (A: RY), frequency (f: RY+CNOT), phase (φ: RY+CNOT) — decoded via analytical inverses.
(3) **Hardware Loop**: circuit arraying (3 circuits + Bell anchor per job), live fidelity tracking (c-indicator), quota-aware submission, timeout handling.

We augment this core with independently validated modules:

- **QDS** (Quantum Delayed Synchronization): Hahn-echo T_2 sweep on `ibm_kingston` qubit 91, 5 runs, raw counts.
- **Quantum Anchor**: post-pulse protocol for coherent-state anchoring on IBM/IQM.
- **Wave-Quantum**: frequency-to-correlation encoding, X-basis ZZ theorem, hardware-validated no-go.
- **QAM** (Quantum Anchored Memory): $\psi(t) = Ae^{-\gamma t}\cos(2\pi f t + \varphi)$ dynamics fused with Microscope Memory (9-level zoom D0–D8, 13-layer consciousness, mmap, SIMD, spine bridge REST/MCP).
- **scs-quantum**: TKR frequency-domain wave-packet encoding, Bell balance 99.3%.

This paper reports the *complete measured record*: 108 QPU iterations on `ibm_fez`, QDS T_2 data, and the architectural synthesis into QAM. All raw data, scripts, and crates are version-controlled and reproducible.

2 RELATED WORK

Quantum-classical loops. Variational algorithms (VQE, QAOA) iterate parameter updates classically [3, 8]. Quantum kernels embed data in Hilbert space [5]. Our loop differs: the quantum subroutine *measures* a wavepacket (A, f, φ) injected by classical dynamics, and the decoded values *perturb* the fluid — a dynamical coupling, not optimization.

Gross-Pitaevskii in cognition. GPE models appear in Bose-Einstein condensates [9] and neural field theories [1]. Fluid STRATOS implements 16 modes on a 2D lattice with conserved density (max $|\Delta| = 2.766$, coherence 0.996 after 1000 steps [10]).

Hardware-validated quantum memory. Quantum RAM proposals [4] remain theoretical. QAM grounds $\psi(t)$ in measured T_2 and anchor protocols, with classical Microscope Memory providing durable, queryable storage.

Error mitigation. Zero-noise extrapolation [11], probabilistic error cancellation [2], and measurement calibration [7] address symmetric noise. Our c-indicator exposes *asymmetric* readout error (floor/ceiling at $P \approx 0.037/0.93$) and bimodal φ dynamics, limiting correction efficacy.

3 SYSTEM ARCHITECTURE

3.1 Fluid STRATOS (GPE Solver)

The 2D GPE on $N \times N$ grid ($N = 64$ default):

$$i\frac{\partial \psi}{\partial t} = -\frac{1}{2}\nabla^2\psi + g|\psi|^2\psi + V\psi \tag{1}$$

Strang splitting: $K/2 \to V \to K/2$ with cached FFT plans. Kinetic step in Fourier space:

$$\psi(\mathbf{k}, t + \Delta t/2) = \psi(\mathbf{k}, t) \exp\left(-i\frac{|\mathbf{k}|^2}{2}\frac{\Delta t}{2}\right) \tag{2}$$

Thread-local scratch buffers (`RefCell<Vec<Complex<f64>>>`) avoid allocation in parallel region. 160 regression tests pass; clippy clean.

3.2 Quantum Bridge Encoding

Three independent circuits per iteration (2 qubits, 1024 shots each):

Table 1: Encoding circuits. $\theta_A = A$, $\theta_f = 2\pi f$, $\theta_\varphi = \varphi$.

Channel	Qubits	Circuit	Ideal $P(q_0 = 1)$
A (amplitude)	1	$\mathrm{RY}(\theta_A)$	$\sin^2(A/2)$
f (frequency)	2	$\mathrm{RY}(\theta_f) \to \mathrm{CNOT}$	$\sin^2(\pi f)$
φ (phase)	2	$\mathrm{RY}(\theta_\varphi) \to \mathrm{CNOT}$	$\sin^2(\varphi/2)$

Decoding (analytical, implemented in Rust `bridge-cli`):

$$A = 2\arccos(\sqrt{P(q_0 = 0)}) \tag{3}$$

$$\varphi = 2\arcsin(\sqrt{P(q_0 = 1)}) \tag{4}$$

$$f = \arcsin(\sqrt{P(q_0 = 1)})/\pi \tag{5}$$

No RZ gates (virtual Z not calibrated on open plan). Degradation model for simulation:

$$p_{\text{meas}} = c \cdot p_{\text{ideal}} + (1 - c) \cdot 0.5 \tag{6}$$

with inverse `unbias_probability` (clamped to $[0, 1]$). 18/18 tests pass.

3.3 Hardware Loop Protocol

Algorithm 1 Quantum-Classical Hardware Loop

```
ψ_0 ← FluidStratos.init()
for i = 1 → N_iter do
    Extract (A_i, f_i, φ_i) from dominant mode of ψ_{i−1}
    Submit job: 3 circuits (A_i, f_i, φ_i) + Bell anchor (if i mod 5 = 0)
    Wait for job completion (timeout 900 s)
    Decode (A'_i, f'_i, φ'_i) via bridge-cli using live c
    ψ_i ← FluidStratos.tick(ψ_{i−1}, A'_i, f'_i, φ'_i)
    if i mod 5 = 0 then
        Update c ← Bell fidelity from anchor
    end if
end for
```

Circuit arraying bundles 4 circuits/job (3 signal + 1 Bell), reducing queue overhead. Pre-submission quota check via `service.usage()` avoids silent queue-stuck (exhausted quota accepts job but never runs). Token via `-token-file` (JSON), never CLI/env in repo.

4 HARDWARE MEASUREMENT CAMPAIGN

4.1 Platform Constraints

IBM Quantum open plan (3 accounts: `z89zu`, `SILENT`, `jthr` — separate CRNs, separate 600 s quotas):

- No Runtime Sessions: HTTP 400 code 1352 ("not authorized").
- Job-mode queue wait: 6.8–137.5 s (avg 17.9 s/iter).
- QPU time: 2.96 s/iter (3 circuits + anchor × 1024 shots).
- 1000 iterations ≈ 2960 s QPU > 600 s quota ⇒ open plan infeasible for full protocol.

4.2 Run 1: 68 Iterations (`ibm_fez`, seed 20261027)

Table 2: 68-iteration loop summary.

Metric	Value
Total QPU time	201 s
Total wall time	1214.8 s
Avg queue wait/iter	17.9 s
Avg QPU time/iter	2.96 s
c range	0.9170–0.9531
c mean	0.9350
c drift	+0.0361
Mode diversity	7 modes (#0: 27, #7: 10, #6: 9, #15: 9, #14: 6
Classical control modes	1 (single mode, 1000/1000)
TV distance (quantum vs uniform)	0.2960

Per-channel decoding (mean absolute error over 68 iterations):

Deterministic proof: for ideal measurement ($A_{\text{true}} = 0.3$), c-correction yields error −0.300 (saturation). The c-indicator tracks drift; it is *not* a decoding correction.

Table 3: Decoding accuracy: raw vs c-corrected.

Channel	Raw MAE	c-corrected MAE	Note
A	0.026	0.212	**Correction harms** (asymmetric noise)
f	0.012	0.011	Slight improvement
φ	0.38	0.34	Bimodal phase ($0/\pi$) dominates

4.3 Run 2: 16-Mode Argmax (6 trials, 1024 shots)

Table 4: 16-mode argmax on hardware.

Metric	Quantum	Classical Control
Bell fidelity	93.95%	—
Argmax accuracy	0/6 (0.00%)	6/6 (100%)
Quantum winners	Creativity 4×, Logic 2×	Brain 0×

The 1.85% gap between Brain (0.096210) and Creativity (0.094434) requires ≈248k shots for 3-σ separation (242× QDS 1024). **Conclusion:** 16-mode discrimination via quantum path is not detectable with current hardware parameters.

4.4 Run 3: 20 Iterations Raw Feedback (seed 20261027)

No c-correction, raw decoded values fed back.

Table 5: 20-iteration raw feedback vs classical control.

Metric	Quantum (raw)	Classical
Modes visited	6 (#0, #6, #7)	1 (Brain, 20/20)
Mean A MAE	0.038	—
c-corrected A MAE	0.095	—

Classical control locking to Brain (single mode) confirms the mode diversity originates from quantum feedback, not fluid instability.

4.5 Run 4: Cross-Device Replication

- `ibm_kingston`: 52 jobs queued, 900 s timeout → `RuntimeJobTimeoutError`, job cancelled cleanly (timeout handler validated).
- `ibm_fez` replication (seed 20261027): 20 iterations, 3 modes (#0, #6, #7), avg wait 19.6 s/iter.

4.6 QDS Validation (Quantum Delayed Synchronization)

Hahn-echo T_2 sweep on `ibm_kingston` qubit 91, 5 independent runs, 1024 shots/delay.

Coherence time supports anchor protocol window ($< 100\ \mu$s).

Table 6: QDS T_2 results.

Run	T_2 (μs)	χ^2_{red}
1	92.3	1.04
2	87.1	0.98
3	84.9	1.12
4	94.2	1.01
5	89.5	0.95
Mean ± SD	89.6 ± 39.2	—

4.7 Quantum Anchor Protocol

Post-pulse coherent-state anchoring on IBM/IQM:

(1) Prepare target state via calibrated pulse.
(2) Apply echo sequence at variable delay τ.
(3) Measure overlap with reference; fit exponential decay.
(4) Anchor accepted if fidelity > 0.95 at protocol τ.

Validated on `ibm_fez` and IQM denominators; enables QAM state initialization.

4.8 Wave-Quantum Encoding

Frequency-to-correlation mapping: encode ω in RY angle, measure ZZ correlation in X-basis. Pearson $r = 0.87$–0.999 across frequencies. X-basis ZZ theorem proves no-go for direct frequency discrimination without tomography.

4.9 QAM: Quantum Anchored Memory

QAM fuses three layers:

(1) **Wave dynamics**: $\psi(t) = Ae^{-\gamma t}\cos(2\pi ft+\varphi)$ with hardware-measured (A, f, φ, γ).
(2) **Quantum Anchor**: coherent-state initialization + periodic re-anchoring (fidelity threshold).
(3) **Microscope Memory**: 9-level zoom (D0–D8), 13-layer consciousness architecture, binary mmap, SIMD, spine bus REST/MCP, Merkle integrity. 41/41 gate tests pass (`unified-mind`).

The spine bridge exposes `POST /recall`, `POST /store`, `GET /stats` with zero JSON overhead (binary frames). 926,573 blocks / 208.5 MB deep archive (`D:/codex/microscope-memory/data`) + 5,622 blocks / 1.4 MB live portable (`AppData/Local`).

4.10 scs-quantum: TKR Encoding

Time-Knowledge Representation (TKR): frequency-domain wavepacket encoding on superconducting QPUs. Bell reference 99.3%, GHZ ideal. Provides the encoding foundation for QAM wavepackets.

5 RESULTS SYNTHESIS

6 DISCUSSION

6.1 Asymmetric Readout Noise

The measurement floor ($P_{\min} \approx 0.037$) and ceiling ($P_{\max} \approx 0.93$) create asymmetric error: $p_{\text{meas}} = cp + (1-c)/2$ assumes symmetric depolarization, which fails. The c-indicator remains useful for *monitoring* drift (0.917–0.953), but *correction amplifies error.*

6.2 Bimodal Phase Dynamics

φ clusters near 0 and π (boundary of dynamical domain). At boundaries, readout noise $\pm\delta P$ maps to large $\delta\varphi$ via arcsin nonlinearity.

Table 7: Unified results across all modules.

Module	Key Metric	Value	Status
Fluid STRATOS	Density conservation max $\|\Delta\|$	2.766	Verified (160 tests)
Fluid STRATOS	Coherence (1000 steps)	0.996	Verified
Quantum Bridge	Bell fidelity (hardware)	93.95%	Measured
Quantum Bridge	16-mode argmax accuracy	0.00%	Measured (6 trials)
Quantum Bridge	A raw MAE	0.026	Measured
Quantum Bridge	c-corrected A MAE	0.212	**Worse**
Quantum Bridge	φ MAE	0.34–0.38	Bimodal phase
Hardware Loop	68-iter mode diversity	7 modes	Measured
Hardware Loop	Classical control modes	1 mode	Verified
QDS	T_2 (qubit 91)	89.6 ± 39.2 μs	5 runs
Quantum Anchor	Fidelity threshold	> 0.95	Protocol defined
Wave-Quantum	Pearson r (freq $\rightarrow$ corr)	0.87–0.99	Measured
Microscope Memory	Gate tests	41/41 pass	Verified
Microscope Memory	Deep archive	926k blocks / 208 MB	Operational

Phase unwrapping or alternative encoding (e.g., $\sin\varphi, \cos\varphi$ via 2-qubit tomography) needed.

6.3 16-Mode Discrimination Gap

The Brain vs. Creativity energy gap (1.85%) requires ~248k shots for 3-σ statistical separation — 242× current 1024. On open plan: 248k shots × 2.96 s/1024 ≈ 717 s QPU > 600 s quota. **Fundamental limit:** open plan cannot resolve the mode competition quantum-mechanically.

6.4 Quantum Noise as Perturbation, Not Structure

Simulation (numpy FFT, 4 branches): quantum-loop $H/H_{\max} = 0.4355$ (top mode 71.4% last 500) vs classical 0.0000 (1000/1000 single mode) vs uniform 0.9962. TV distance 0.2960. Quantum noise alone is unstructured; GPE provides attractor; quantum kick stabilizes/destabilizes.

6.5 Biological Plausibility of Reconsolidation

QAM's $\psi(t)$ decay + anchor reconsolidation mirrors memory reconsolidation windows [6]. The 9-level zoom (D0–D8) maps to hippocampal-cortical hierarchy. Cross-validation invited.

6.6 Open Plan vs. Dedicated Session

Open plan: no sessions, job queue measured, quota hard limit. Dedicated/flex plan: session reservation, lower latency, higher quota. **1000-iteration protocol requires dedicated plan.**

7 REPRODUCIBILITY & ARTIFACTS

- **Repository**: https://github.com/silentnoisehun/quantum-organic-intelligence (public)
- **Raw data**: `measurement_raw/` (104 JSON files, job IDs, timestamps, version-controlled)
- **Reports**: `docs/MEASUREMENTS-HARDWARE-20261027.txt`, `RUNLOG-*.txt`
- **Summary script**: `scripts/summarize_hardware_run.py` (runs in CI)
- **Rust crates**: `fluid-stratos` (12 tests), `quantum-bridge` (18 tests, `bridge-cli` binary)
- **Python env**: `.qoi-venv` (qiskit 2.6.1, qiskit-ibm-runtime 0.51.0, numpy 2.5.4)
- **Commands**:

```
# Local validation (no token)
python scripts/hardware_measure.py --local --shots 256 --mode-

# Hardware (requires IBM_QUANTUM_API_TOKEN)
python scripts/hardware_measure.py --backend ibm_fez --shots 1
python scripts/quantum_loop_hardware.py --backend ibm_fez --it
```

- **CI**: 4 consecutive green runs (fmt, clippy, tests, measurement scripts)

8 CONCLUSION

We built and measured a complete quantum-classical feedback system on IBM superconducting hardware. The QOI loop runs (108 iterations documented), but the quantum path does not outperform classical control for mode discrimination (0% vs 100%). The c-indicator tracks hardware drift but fails as correction due to asymmetric readout noise. Phase bimodality dominates φ error. The 16-mode gap is statistically unresolved on open-plan quota.

Complementary modules (QDS, Quantum Anchor, Wave-Quantum, QAM, Microscope Memory, scs-quantum) provide independently validated building blocks for a neuromorphic quantum memory architecture. The full stack is open: code, data, measurements, and limits.

Future work: dedicated IBM plan for 1000 iterations; asymmetric noise model for mitigation; phase-unwrapped encoding; QAM biological cross-validation; IQM denominator comparison.

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A APPENDIX A: REPOSITORY INDEX

Table 8: All 7 repositories in the QOI ecosystem.

Repository	Focus	Key Result
quantum-organic-intelligence	GPE + Bridge + Loop	108 hardware iterations
quantum-anchor	Coherent-state anchoring	Protocol + IBM/IQM validation
wave-quantum	Freq-to-correlation encoding	Pearson 0.87–0.999, ZZ no-go
quantum-delayed-sync	Hahn-echo T_2 sweep	89.6 ± 39.2 μs (5 runs)
quantum-anchored-memory	$\psi(t)$ + Memory + Anchor	QAM architecture
scs-quantum	TKR wave-packet encoding	Bell 99.3%, GHZ ideal
microscope-memory	9-level cognitive memory	41/41 gates, 208 MB archive

B APPENDIX B: HARDWARE LOOP PARAMETERS

```
Backend: ibm_fez (156 qubit, active)
Shots: 1024 per circuit
Circuits/iter: 3 (A, f, phi) + Bell anchor every 5
Anchor update: live c fed to bridge-cli decoder
Timeout: 900 s per job (clean cancel on expiry)
Quota guard: pre-submission service.usage() check
Token: --token-file (JSON), never in repo/env
Venv: qoi_venv (qiskit>=2.0, qiskit-ibm-runtime>=0.50)
```

C APPENDIX C: CLASSICAL CONTROL BASELINE

Fluid SHRATOS without quantum feedback converges to single attractor (brain) in all 1000/1000 trials. This establishes that mode diversity in quantum runs originates from the quantum feedback channel, not fluid instability.